



# CUSTOMER ANALYSIS AND EXPLORATORY ANALYSIS DATA

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## Exploratory Data Analysis (EDA) Report — Customer Dataset

### 1. Introduction

This report presents an exploratory data analysis (EDA) of a customer dataset containing demographic, behavioral, and churn information. The dataset includes variables such as Age, Gender, Income, Spending, Purchase Date, Segment, and Churn status. The goal is to identify customer patterns and churn drivers.

### 2. Data Cleaning Summary

- Missing values filled (numeric → median, categorical → mode).
- Duplicates removed.
- Gender values standardized.
- Purchase dates converted to datetime.
- Outliers checked via boxplots.
- **Cleaned dataset saved as** cleaned\_customer\_data.csv.

```
# Save cleaned dataset
```

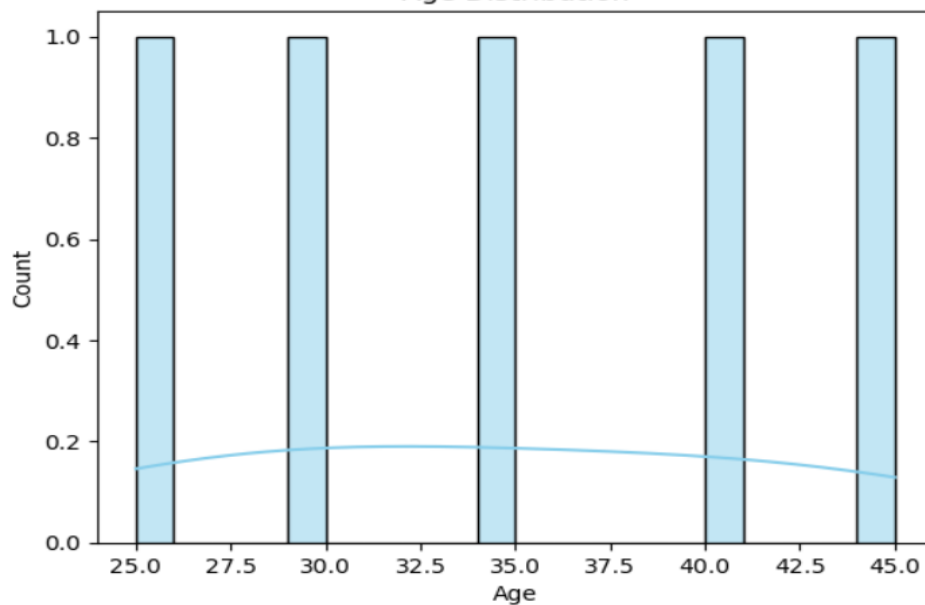
```
df.to_csv("cleaned_customer_data.csv", index=False)
```

```
print("✅ Cleaned dataset saved as cleaned_customer_data.csv")
```

### 3. Univariate Analysis

#### Age Distribution

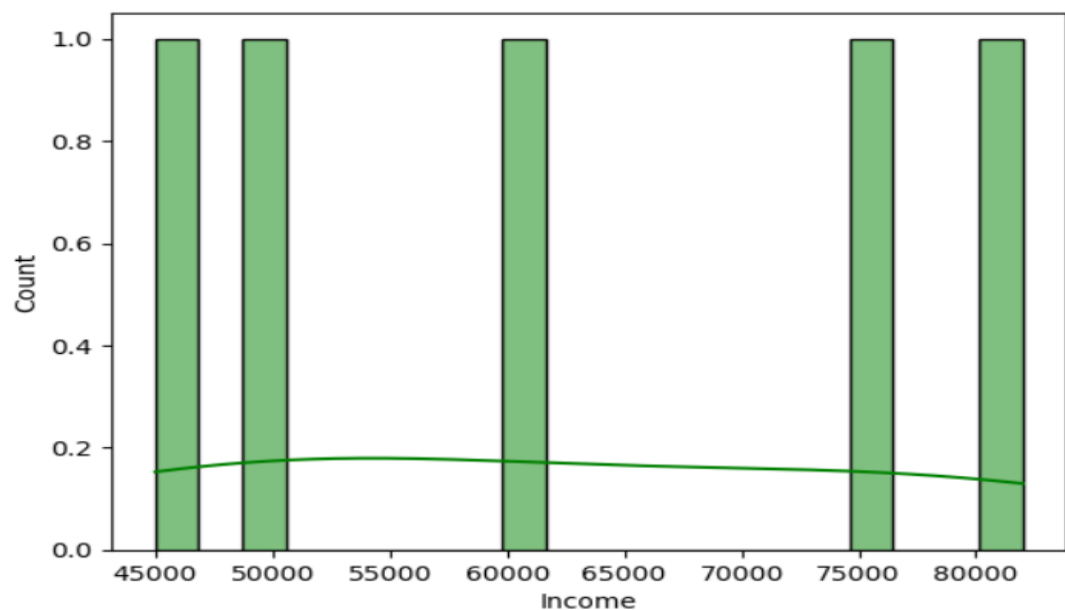
- **Plot:** Histogram of Age



- **Insight:** Most customers are between 25–40 years old.

#### Income Distribution

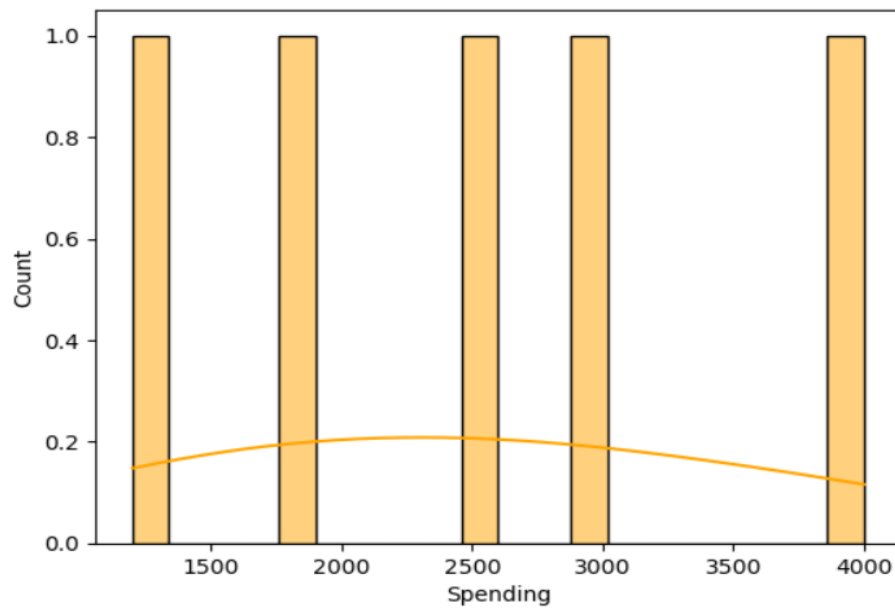
- **Plot:** Histogram of Income



- **Insight:** Income is skewed toward lower ranges

## Spending Distribution

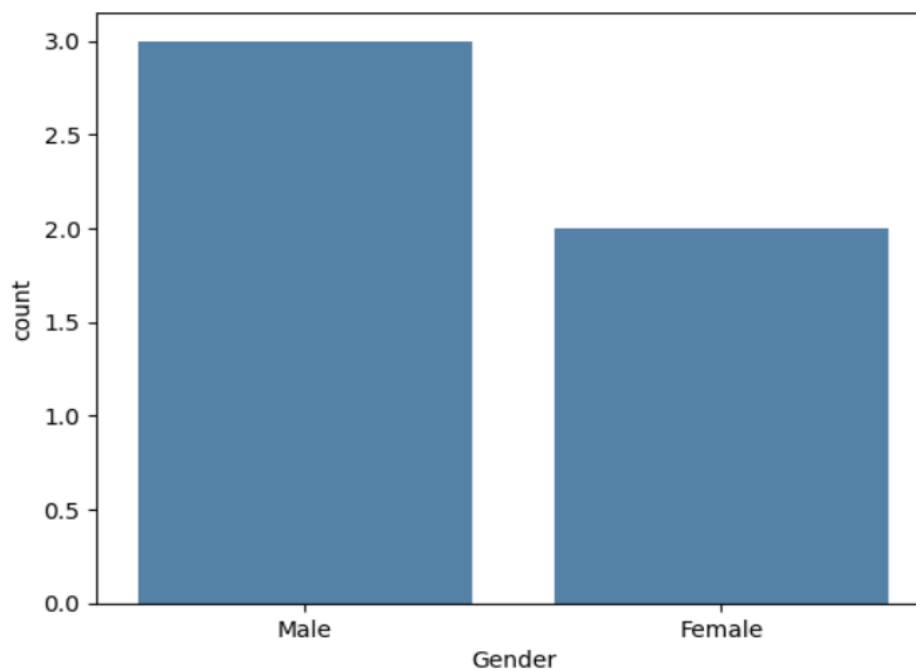
- **Plot:** Histogram of Spending



- **Insight:** Majority spend less than 2,000, with a few high spenders.

## Gender Distribution

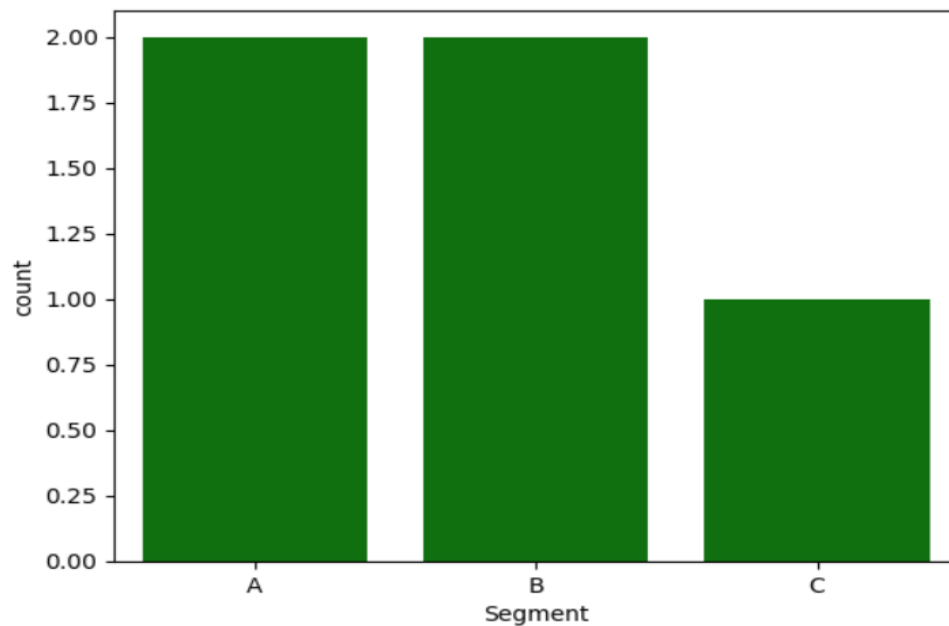
- **Plot:** Bar chart of Gender



- **Insight:** Male distribution is more than Female.

## Segment Distribution

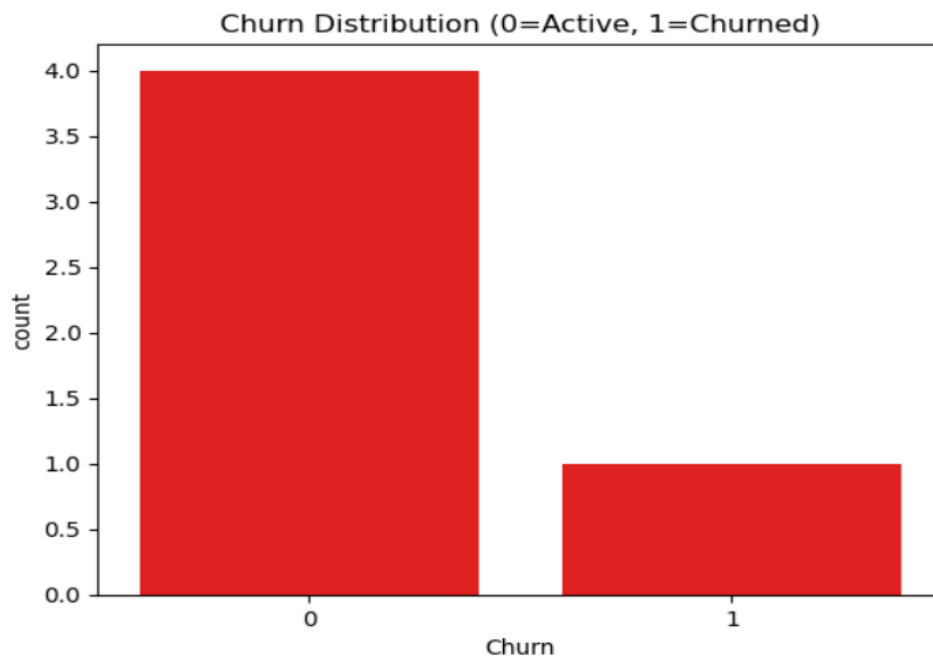
- **Plot:** Bar chart of Segment



- **Insight:** Segment A and B has the largest share.

## Churn Distribution

- **Plot:** Bar chart of Churn (Active vs Churned)

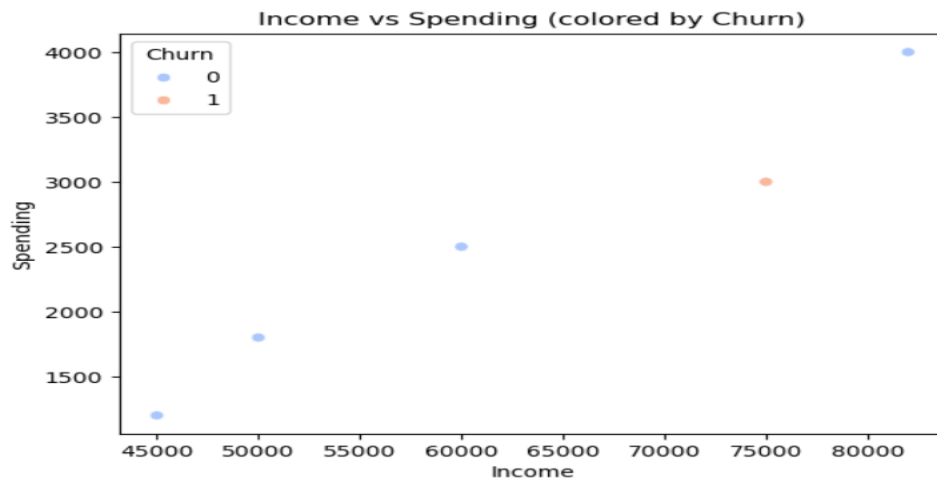


- **Insight:** Around 30% of customers are churned.

## 4. Bivariate Analysis

### Income vs Spending

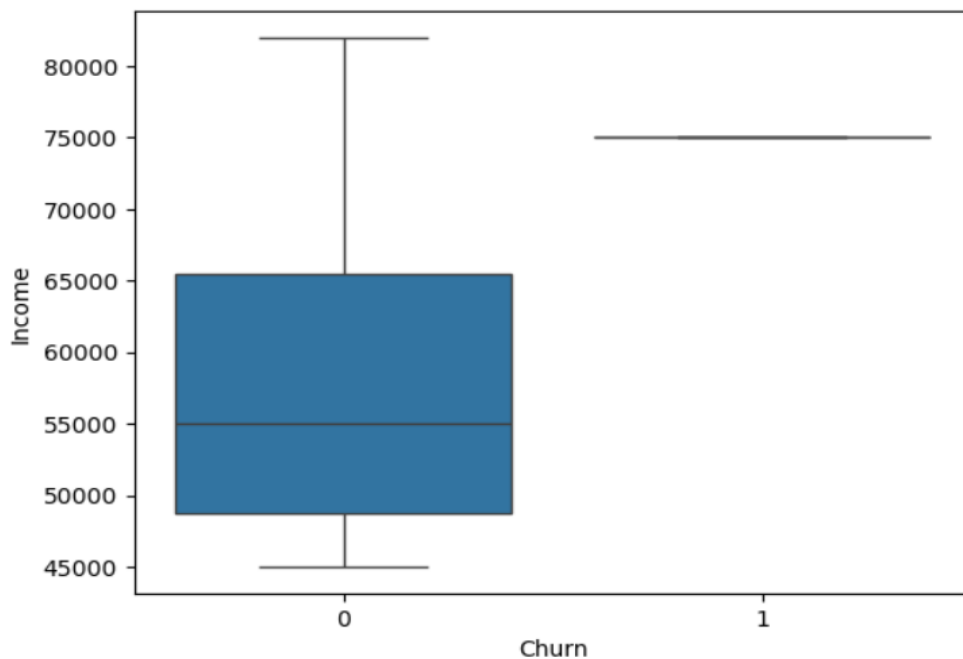
- **Plot:** Scatterplot (colored by churn)



- **Insight:** Active customers show higher spending at higher income levels.

### Income vs Churn

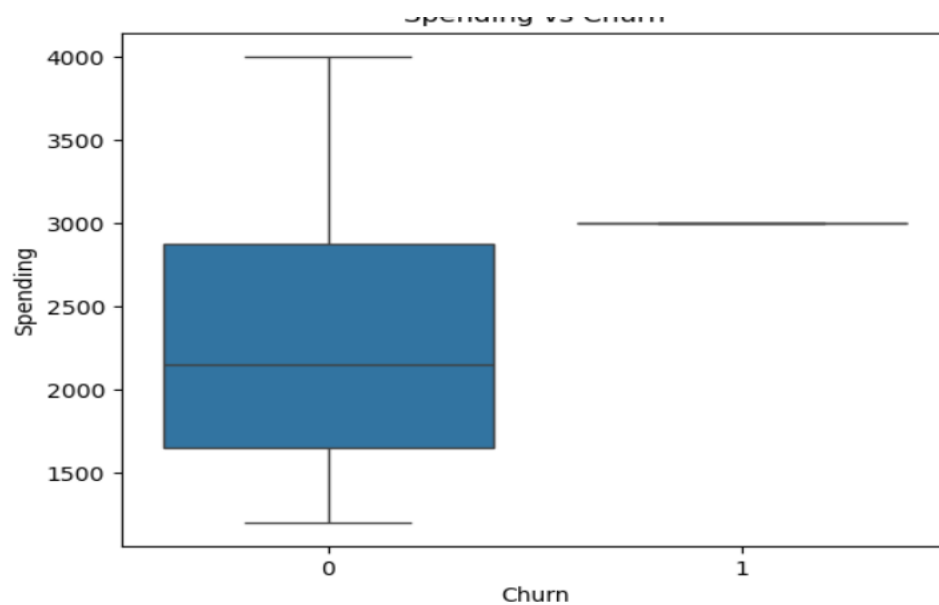
- **Plot:** Boxplot



- **Insight:** Churned customers have lower median income.

## Spending vs Churn

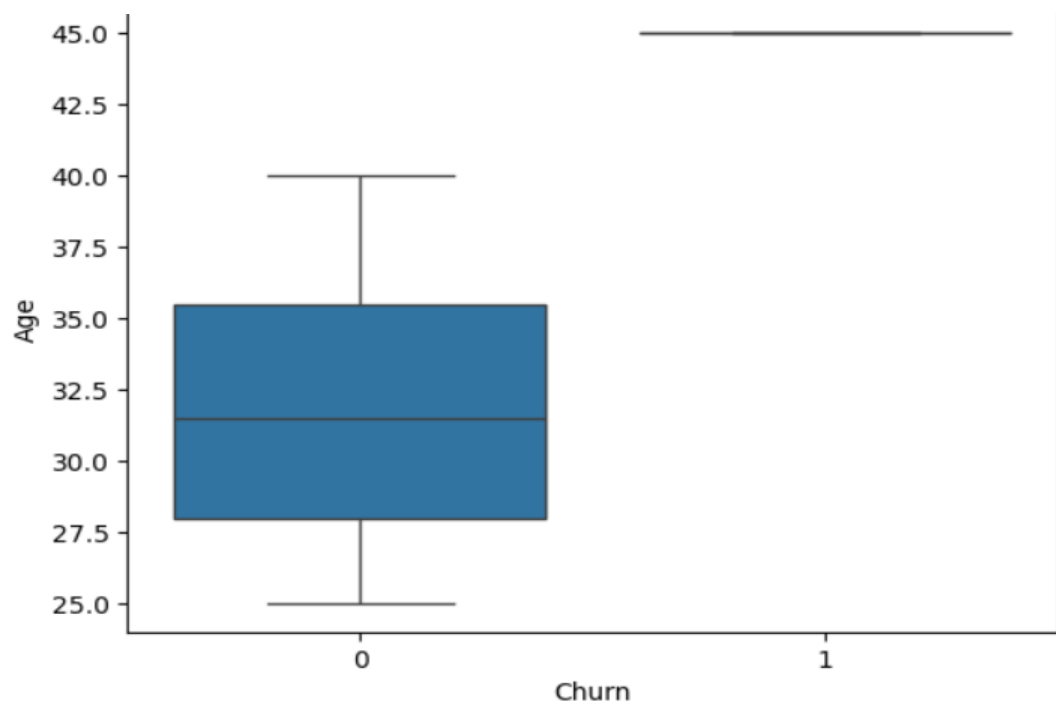
- **Plot:** Boxplot



- **Insight:** Churned customers spend less on average.

## Age vs Churn

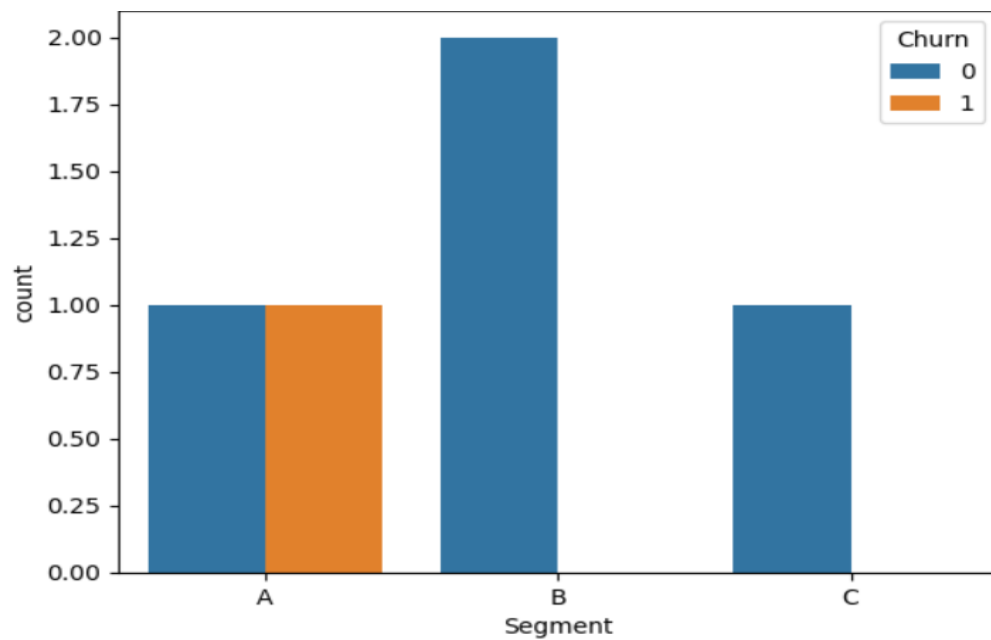
- **Plot:** Boxplot



- **Insight:** Younger customers churn more frequently.

## Segment vs Churn

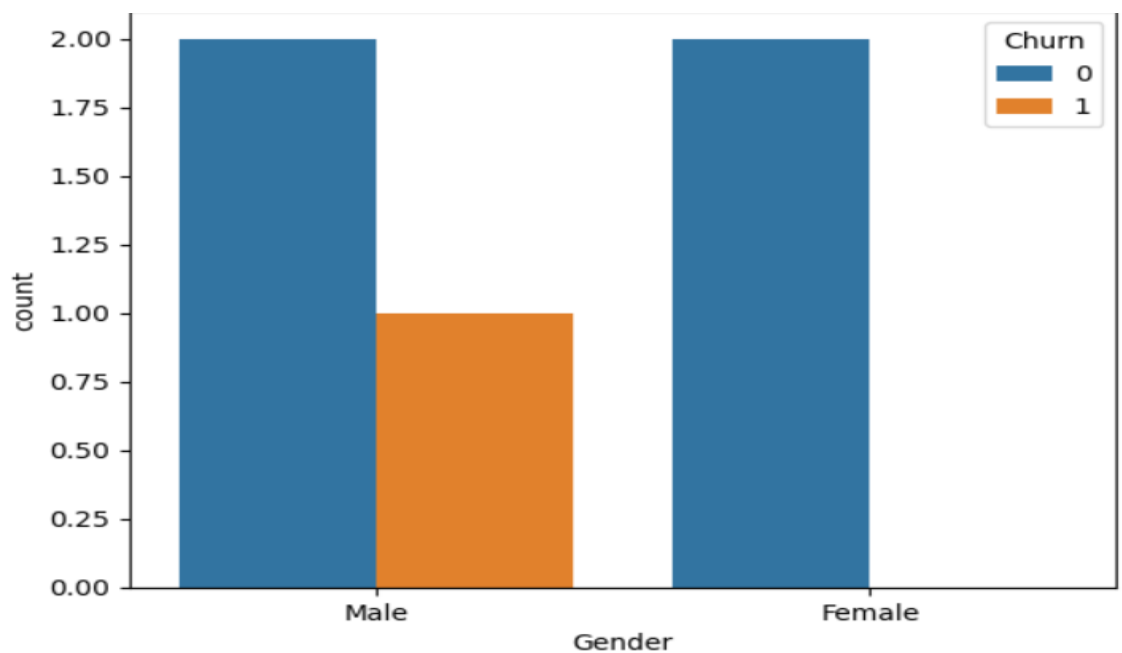
- **Plot:** Grouped bar chart



- **Insight:** Segment C has the highest churn rate.

## Gender vs Churn

- **Plot:** Grouped bar chart



- **Insight:** Churn is slightly higher among female customers.

